

Magnitude-Weighted Signed Rayleigh Cancellation in Finite Prime Shells

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Abstract

The preceding signed-shell study treated every nonzero Gram cross term as a unit edge. That sign layer can identify compatible cuts, but it does not measure the physical quadratic form. This paper restores the exact Gram magnitudes and studies the trace-normalized quotient over equal coefficient signs. We prove the diagonal-plus-cross-term identity, prove that a common-denominator exhaustive enumeration is globally optimal modulo global sign reversal, and certify the resulting finite atlas with exact rational arithmetic. On the inherited 18-row literal grid, all 18 globally optimized quotients are below one, all 18 all-positive quotients are above one, and the weighted optimizer differs from the unit-edge max-cut optimizer on all 18 rows. The result is a finite weighted structural diagnostic: source-image feasibility, growing-shell control, arithmetic L^2 bounds, and the twin-prime endpoint remain open.

1 Position on the twin-prime route

This release stays inside the same literal prime-shell dynamical family. The earlier papers separated physical prime components and measured their Gram coherence [4, 3, 2]. Adaptive nonnegative weighting found a coherence wall [1]; a signed pair direction was then isolated in TPC-291 [6]. TPC-292 showed that three pairwise preferences obey a cycle-parity obstruction [7], and TPC-293 lifted that sign question to a complete signed shell graph [5]. The present paper performs the smallest missing test: put the Gram magnitudes back into the objective.

The distinction is essential. A unit-weight max-cut label optimizes a combinatorial surrogate. The physical norm weights each cross term by its actual magnitude and also carries a fixed diagonal contribution. We compare the two optimizers directly rather than identifying them by assumption.

2 Physical shell

Let $I = I_N$ be the frozen integer interval and let β be the frozen rational source vector. For an odd prime q , height H , and exponent s , write

$$K_{H,s}(h) = \frac{H^{2s}}{(H^2 + h^2)^s}, \quad B_q(u, t) = \mathbf{1}_{q \nmid u} \mathbf{1}_{q \nmid t} \left(\mathbf{1}_{u \equiv t \pmod{q}} - \frac{1}{q-1} \right).$$

The deleted-diagonal physical component is $g_q = A_q \beta$, where

$$(A_q)_{u,t} = q K_{H,s}(u-t) \mathbf{1}_{u \neq t} B_q(u, t).$$

For a finite prime shell $S = \{q : Q < q \leq 2Q\}$, define the exact Gram matrix

$$G_{q,r} = \langle g_q, g_r \rangle, \quad \text{tr } G = \sum_{q \in S} G_{q,q}.$$

All quantities on the declared grid are rational.

3 Weighted sign quotient

For an equal-coefficient sign vector $a \in \{-1, +1\}^S$, define

$$R(a) = \frac{a^\top G a}{\text{tr } G}.$$

The diagonal part is fixed, and symmetry gives the following exact identity.

Theorem 1 (Trace-normalized signed quadratic identity). *If G is symmetric with positive trace, then*

$$R(a) = 1 + \frac{2 \sum_{q < r} a_q a_r G_{q,r}}{\text{tr } G}.$$

Moreover $R(a) \geq 0$ when G is a Gram matrix.

Proof. Expand $a^\top G a$. Since $a_q^2 = 1$, its diagonal contribution is $\sum_q G_{q,q} = \text{tr } G$, and symmetry combines the two copies of each upper-triangular cross term. This proves (1). If G is a Gram matrix, then $a^\top G a = \left\| \sum_q a_q g_q \right\|^2 \geq 0$. $\square$

The unit-edge objective of TPC-293 keeps only the signs of $G_{q,r}$. In contrast, (1) retains the weights $|G_{q,r}|$ and minimizes their signed sum. This is the precise layer at which the two routes can separate.

4 Exact finite optimization

Let D be the least common multiple of all denominators in G and put $M = DG$. Then M is an integer symmetric matrix and

$$R(a) = \frac{a^\top M a}{\text{tr } M}.$$

Global reversal $a \mapsto -a$ leaves this value unchanged. Thus fixing one coordinate, say $a_0 = +1$, leaves exactly $2^{|S|-1}$ representatives.

Theorem 2 (Global finite sign search). *For rational G with positive trace, exhaustive enumeration of those $2^{|S|-1}$ representatives returns the global minimum and maximum of $R(a)$ over all sign vectors.*

Proof. Multiplication by the positive D preserves ordering. Every sign vector is either represented by exactly one vector with first coordinate $+1$ or by its global reversal, and the quadratic form is invariant under that reversal. The restricted enumeration therefore contains one representative of every equivalence class, so its extrema are the unrestricted extrema. $\square$

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For the producer, a Gray traversal updates an integer quadratic value after a single label flip. If $F_v = \sum_{j \neq v} M_{v,j} a_j$ and the old label is a_v , then

$$(a')^\top M a - a^\top M a = -4a_v F_v, \text{quad} F'_u - F_u = -2a_v M_{u,v} \quad (u \neq v).$$

The independent checker uses direct reflected-binary enumeration and a source-coordinate-first physical accumulation, so the certificate does not depend on one traversal or one summation order.

5 Finite protocol and atlas

The experiment inherits eight $s = 2$ growth rows, four $s = 1$ exponent crossover rows, and six $s = 2$ source-control rows. For every row, all primes in $Q < q \leq 2Q$ are evaluated exactly, their Gram matrix is integerized, and every equal-sign labeling is visited. The unit-edge TPC-293 max-cut witness is also evaluated against the weighted matrix.

Table 2 gives representative exact-rational results. Decimal values are display-only; all comparisons in the certificate use fractions.

The smallest quotient in the grid is 0.0496374497659 at $(N, H, Q, z, s) = (512, 58, 90, 5, 2)$, while the largest all-positive quotient is 16.4393131994 at the same row. The comparison is especially informative at the TPC-293 exceptional row (256, 38, 27, 5, 1). Its unit-edge max-cut label gives $R = 0.988974603760$, but the exact weighted optimum gives $R = 0.519059163428$. Across the full grid, the weighted optimum differs from the max-cut value in every row.

Table 1: Global finite audit on the inherited grid.

quantity	count
rows	18
shell edges	1,380
weighted minima below one	18 / 18
all-positive quotients above one	18 / 18
max-cut candidate quotients below one	18 / 18
weighted optimum differs from max-cut	18 / 18
weighted optimum at most 1/4	13 / 18
weighted optimum at most 1/10	8 / 18

Table 2: Representative trace-normalized quotients.

axis	N	H	Q	s	$ S $	R_+	$R_{\min}$
growth	128	24	9	2	3	2.125	0.534
growth	192	32	16	2	5	3.444	0.409
growth	384	50	40	2	10	7.107	0.182
growth	512	58	60	2	13	10.993	0.108
growth	512	58	90	2	17	16.439	0.0496
crossover	256	38	27	1	7	2.167	0.519
crossover	512	58	90	1	17	12.363	0.143
control	384	52	70	2	15	14.107	0.0638

6 Interpretation and route boundary

The finite result closes a genuine missing branch of the map. The sign-frustration atlas did not by itself say whether a favorable edge pattern carried enough mass. TPC-294 answers that finite question affirmatively for the ambient equal-sign cube, while simultaneously showing that unit-edge max-cut is not the right physical optimizer.

There is a second, equally important boundary. The signs a_q were optimized in the ambient coefficient-sign cube; they were not required to be images of a coefficient vector generated by the frozen source map. The source-native image may be much smaller, and no surjectivity or approximation theorem is proved here. Nor does an 18-row atlas imply a growing-shell estimate. Thus the result pays no fixed-power credit and leaves arithmetic L^2 , Gate B, and the twin-prime endpoint open.

The next natural experiment is not another unconstrained sign search. It is an image audit: compute the coefficient/sign patterns induced by admissible source perturbations, compare them with the weighted minimizers, and measure the least attainable weighted quotient. This is recorded as the `ROUND2_CLUE` in the certificate.

7 Reproducibility and conclusion

From the project directory, run:

```
export PYTHONDONTWRITEBYTECODE=1
python -B code/tpc294_magnitude_weighted_signed_rayleigh_certificate.py --check
python -B experiments/tpc294_independent_checker.py
python -B experiments/tpc294_magnitude_weighted_stress.py
```

The canonical result is `results/tpc294_certificate.json`. Normal and optimized invocations are required to have empty standard error and byte-identical standard output.

The Session-named Route-A/Route-B evaluator files are absent from this checkout; the proof package, exact certificate, independent replay, stress test, and Bridge-B checker provide the available local fail-closed evidence.

References

- [1] Liang Wang. Adaptive shell weighting and the nonnegative coherence wall, 2026. TPC-290 project release in the same research repository.
- [2] Liang Wang. Cross-prime gram coherence in a literal twin-prime dynamics, 2026. TPC-289 project release in the same research repository.

- [3] Liang Wang. Growing-shell gram obstruction in a literal twin-prime dynamics, 2026. TPC-288 project release in the same research repository.
- [4] Liang Wang. Prime-shell cancellation depth in a literal twin-prime dynamics, 2026. TPC-287 project release in the same research repository.
- [5] Liang Wang. A signed max-cut obstruction for multi-prime cancellation, 2026. TPC-293 project release in the same research repository.
- [6] Liang Wang. Signed schur cancellation in cross-prime gram pairs, 2026. TPC-291 project release in the same research repository.
- [7] Liang Wang. Three-prime sign frustration in a physical gram shell, 2026. TPC-292 project release in the same research repository.